

Reaction Time Combine

Train like a sports scientist and prove which strategy cuts delay.

Win condition: Best median plus most improvement using clean data.

THE SCIENCE YOU ARE USING

From signal to muscle. Reaction time is the delay between seeing a signal and moving. The eye sends a signal to the brain, the brain decides, and nerves fire the muscle. The classic ruler-drop test turns that delay into a distance you can measure.

Median and improvement. One trial is noisy, so you take many and use the median. Then you test whether a strategy, like focusing or warming up, actually improves your number. Clean data and real improvement both score.

HOW TO BUILD AND RUN IT

- 1 Run the ruler drop.** A partner drops a meter stick; you catch it. The catch distance maps to your reaction time.
- 2 Take many trials.** Record at least ten catches. Use the median, not the best single catch.
- 3 Pick a strategy.** Choose one strategy to test: focus cue, warm-up, or eyes on the release point.
- 4 Retest and compare.** Run another set with the strategy. Compare medians to see if it helped.
- 5 Show the data.** Present a clear before-and-after with your median and spread.

Safety: Clear lanes. No collisions. All stations can be seated. Allergy: None.

MATERIALS

Meter sticks	per team
Stopwatches	per team
Cones	shared
Score sheets	per team
Clipboards	6

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Best median reaction time	35
Data quality	20
Strategy improvement	20
Explanation	15
Teamwork	10
Total	100